Customer Support Assistant (NLP + RL)

Rafael Allan Alvarado
rdalvarado@student.hau.edu.ph

Louis Gabriel Malig
lvmalig@student.hau.edu.ph

Redenelle Maurice Manangu
rlmanangu@student.hau.edu.ph

Ghayzelle Dizon
gbdizon@student.hau.edu.ph

WEEK 2 PROGRESS REPORT

• Data acquired/cleaned; splits finalized; EDA notebook.

Data acquired/cleaned; splits finalized; EDA notebook. The dataset used for this project is the Bitext Customer Support LLM Chatbot Training Dataset, sourced from HuggingFace (CC BY 4.0). The raw data was cleaned using src/data_pipeline.py, which handled removing null values, stripping duplicate entries, normalizing text to lowercase, and filtering out entries with empty or unusually long texts. After cleaning, the final dataset contains 24,185 labeled utterances across 27 e-commerce intent categories. A stratified 80/10/10 split was applied with a fixed seed of 42, resulting in 19,348 training samples, 2,418 validation samples, and 2,419 test samples. The splits were saved to the output/ folder as separate CSV files to ensure consistency across all training runs. An EDA notebook (notebooks/01_eda.ipynb) was also completed, covering intent distribution across all 27 classes, average utterance length (8.86 words), and a notable class imbalance where cancel_order had significantly fewer samples (~428) compared to other intents (~990+), which was flagged as a potential fairness concern.

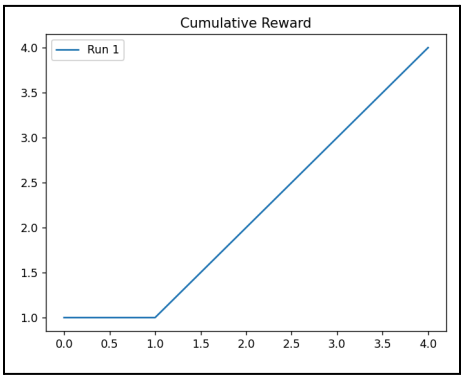
• Draft Model Card sections started; Ethics statement updated.

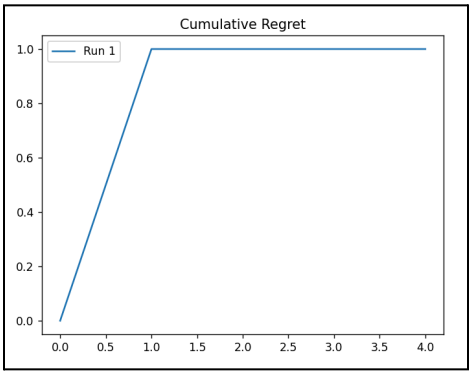
Draft Model Card sections started; Ethics statement updated. A draft model card has been started in docs/model_card.md covering the core model architecture, dataset details, training configuration, intended use, and known limitations. Final metrics are not yet included as full training is still ongoing and will be completed in Week 3. The ethics statement in docs/ethics_statement.md has been written, identifying three main risks associated with the system along with their corresponding mitigations. One mitigation is already reflected in the current codebase, while the remaining ones are planned to be implemented and verified during the evaluation phase in Week 3. A dataset limitation identified during EDA has also been noted and documented in the model card as a known concern going into the final training run.

- **Baselines trained, initial metrics logged. NLP component scaffolded/prototyped.**

The NLP component (DistilBERT) has been fully trained and is confirmed working as the core intent classifier. The model was fine-tuned on the training split using AdamW optimizer with a learning rate of 2e-5, batch size of 16, and early stopping at patience of 2 with macro-F1 as the primary tracked metric. It achieved a macro-F1 of 1.00 and an overall accuracy of 1.00 on the test set across all 27 intent categories, with the confusion matrix showing near-perfect diagonal classification and only minor misclassifications observed in a few classes such as cancel_order (recall: 0.95) and delete_account (recall: 0.98). These results serve as the DL baseline for the project.

- **RL agent stubbed with reward design; early learning curves (even if noisy).**

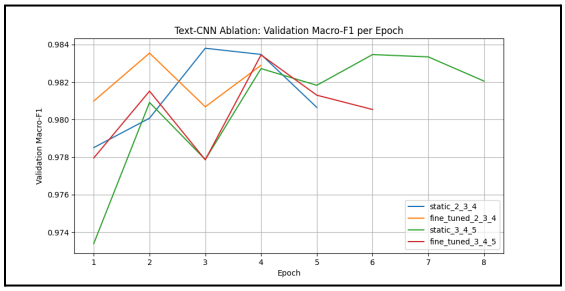




The RL agent has been stubbed and is functional, with the reward design defined as +1 for a correct response

selection and 0 otherwise, using a confidence threshold of 0.70 and an epsilon of 0.10 for exploration. Early learning curves have been generated from an initial interactive run, showing the cumulative reward increasing steadily from 1.0 to 4.0 across 4 steps, while the cumulative regret plateaus at 1.0 after the first incorrect action and remains flat — indicating the agent is consistently selecting correct actions after the initial misclassification

- **CNN experiment running (first results);**



The TextCNN was run across four ablation configurations using GloVe embeddings (400,000 words) static and fine-tuned embeddings with filter sizes [2,3,4] and [3,4,5]. All four runs triggered early stopping and performed closely, with validation macro-F1 scores converging around 0.982–0.984 as shown in the ablation plot. The best performing configuration was fine_tuned_3_4_5 with a test macro-F1 of 0.9839 and accuracy of 0.98, confirming that the CNN component is working well and serving as a strong comparison point against the DistilBERT model. However, it is worth noting that while the CNN performs strongly on the synthetic test set, it may struggle with real-world conversational inputs since it relies on fixed word embeddings and local pattern matching rather than deep contextual understanding, as seen in the interactive test where more informal or ambiguous phrasings produced noticeably lower confidence scores.